Siyuan (Wilson) Sun

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Education

University of Arizona Tucson, AZ
B.S. in Artificial Intelligence; GPA: 3.85/4.0 *Aug. 2025 – May 2027 (Expected)*
- **Relevant Coursework:** Deep Learning for NLP, Neural Networks, Reinforcement Learning, Principles of ML, Text Retrieval and Web Search.

East China University of Science and Technology Shanghai, China
B.S. in Artificial Intelligence (Transferred to UArizona) *Sep. 2023 – July 2025*

Research Experience

LUMIA Lab, Shanghai Jiao Tong University Shanghai, China
Research Intern (Advised by Prof. Zhouhan Lin) *Apr. 2026 – Present*
- Researching **parametric memory for LLMs**, LLM pretraining, and test-time training (TTT).

CLULab, University of Arizona Tucson, AZ
Undergraduate Research Assistant (Advised by Prof. Mihai Surdeanu) *Sep. 2025 – Present*
- Researching **LLMs and Information Retrieval**, with a focus on Transformer-based neural retrieval systems and query expansion integration.

Technical Skills

Programming: Python, C/C++, Java, Bash.
Deep Learning & LLMs: PyTorch, Hugging Face Transformers, Accelerate, PEFT (LoRA).
Memory & Retrieval Methods: Parametric memory, RAG, dense retrieval, contrastive learning, hard-negative mining, query expansion, hybrid retrieval, reranking.
Retrieval Tools: FAISS, Pyserini, Sentence Transformers, Elasticsearch, Milvus.
Training & Optimization: Large-scale multi-GPU and multi-node distributed training; DDP, FSDP, DeepSpeed ZeRO; mixed-precision training (FP16/BF16).

Publications

Deep Fusion Memory (DFM) | *LLMs, Parametric Memory* ICLR 2027 (Under Review)
- *Deep Fusion Memory: A Chunk-Level Parametric Memory for Large Language Models.* Rubin Wei, Siyuan Sun, Hao Sun, Yuliang Liu, Xiang Shi, Jiarui Wang, Junming Zhang, and Zhouhan Lin, 2026. Learns chunk-level memory and fuses it into a frozen LLM via gated cross-attention; no explicit retrieval at inference.
- Improved average scores over the strongest baseline by **12.71% / 11.19%** on six QA tasks (Mistral-7B / Qwen3-8B); domain adaptation gains of **6.51 / 4.38 points** on Qwen3-4B / 8B.

AnchorQE | *Dense Retrieval, Query Expansion* WSDM 2027 (Under Review)
- *Query Expansion Is More Than Generation: Improving Dense Retrieval through Better Integration.* Siyuan Sun and Mihai Surdeanu, 2026. Separately encodes queries and expansions, then interpolates their vectors using an unsupervised online weight estimate.
- Improved retrieval effectiveness by up to **12.89%** across TREC-DL, LoTTE, and BEIR without task-specific training; online calibration beat development-tuned fixed weights by up to **3.81%**.

Selected Projects

Emotion Classification in Creative Text | *RoBERTa, Domain Adaptation*
- Transferred emotional priors from social media to poetry/lyrics with two-stage training; compared four tuning strategies, with full fine-tuning achieving the best poetry $F1 \approx 0.60$.

Learning Dense Reward Functions in Chess | *Gymnasium, ResNet, IRL*
- Inferred dense rewards from expert chess demonstrations with a custom Gymnasium environment and ResNet reward model; saliency analysis revealed human-like positional patterns.

Brain-to-Text: Decoding Intracortical Speech | *BERT, GPT-2, LoRA* Brain-to-Text '25
- Built a BERT–GPT-2 neural speech decoder using masked phoneme modeling and LoRA, achieving **12.4% WER**; a day-specific adaptation layer handled non-stationary neural signals.